

Exercise 6.1. *Prove or disprove: If $v_0, \dots, v_{n-1} \in V$, then:*

$$\sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \langle v_j, v_k \rangle \geq 0$$

Proof. For any j , we have $\sum_{k=0}^{n-1} \langle v_j, v_k \rangle = \langle v_j, v_0 + \dots + v_{n-1} \rangle$, thus the equation is now $\langle v_0 + \dots + v_{n-1}, v_0 + \dots + v_{n-1} \rangle \geq 0$, which is trivial by definition. $\square$

Exercise 6.2. *Let $S \in \mathcal{L}(V)$. Define $\langle \cdot, \cdot \rangle_1$ by*

$$\langle u, v \rangle_1 = \langle Su, Sv \rangle$$

for all $u, v \in V$. Show that $\langle \cdot, \cdot \rangle_1$ is an inner product over $V \iff S$ is injective.

Proof.

- Try to prove $\langle \cdot, \cdot \rangle_1$ is a inner product and see where we stuck without injective. We find that $\langle v, v \rangle_1 = 0 \iff v = 0$ stuck, as if S is not injective, then we have $\langle u-v, u-v \rangle_1 = \langle S(u-v), S(u-v) \rangle = \langle 0, 0 \rangle = 0$ where $u-v \neq 0$ and $Su = Sv$. Therefore S must be injective.

Although the proof is not *constructive*, it help us to find a constructive proof: Suppose $Su = Sv$, then $\langle u-v, u-v \rangle_1 = \langle Su-Sv, Su-Sv \rangle = 0$, thus $u-v=0$, therefore $u=v$.

- Positivity, Additivity, Homogeneity holds cause S is a linear map and $\langle \cdot, \cdot \rangle$ is an inner product, and Conjugate Symmetry holds cause $\langle \cdot, \cdot \rangle$ is an inner product. For Definiteness, $\langle v, v \rangle_1 = \langle Sv, Sv \rangle = 0$ implies $v=0$ cause S injective, and $\langle 0, 0 \rangle_1 = \langle S0, S0 \rangle = \langle 0, 0 \rangle = 0$.

$\square$

Exercise 6.3.

- Show that $f((a, b), (c, d)) = |ac| + |bd|$ is not an inner product over $\mathbb{R}^2$.
- Show that $f((a, b, c), (x, y, z)) = ax + cz$ is not an inner product over $\mathbb{R}^3$.

Proof.

- $f((1, 1), (1, 1)) = 1 + 1$ and $f((-1, -1), (1, 1)) = 1 + 1$, then $f((1, 1) + (-1, -1), (1, 1)) = f((0, 0), (1, 1)) = 0 + 0 = 0$ while $f((1, 1) + (-1, -1), (1, 1)) = f((1, 1), (1, 1)) + f((-1, -1), (1, 1)) = 2 + 2 = 4$.
- $f((0, 1, 0), (0, 1, 0)) = 0$ but $(0, 1, 0) \neq 0$.

□

Exercise 6.4. Let $T \in \mathcal{L}(V)$ and $\|Tv\| \leq \|v\|$ for all $v \in V$. Show that $T - \sqrt{2}I$ is injective.

Proof. Suppose $T - \sqrt{2}I$ is not injective, then $Tv = \sqrt{2}v$ for some v , then $\|Tv\| = \|\sqrt{2}v\| = |\sqrt{2}| \|v\| \geq \|v\|$. Basically any eigenvalue with absolute value greater than 1 can make it. □

Exercise 6.5. Let V an inner product space over $\mathbb{R}$.

- Show that $\langle u + v, u - v \rangle = \|u\|^2 - \|v\|^2$.
- Show that $u + v \perp u - v$ if $\|u\| = \|v\|$.
- Use last conclusion to show that the diagonal of 菱形 are orthogonal.

Proof.

- $\langle u + v, u - v \rangle = \langle u, u \rangle + \langle v, u \rangle - \langle u, v \rangle - \langle v, v \rangle$, note that $\langle v, u \rangle = \langle u, v \rangle$ since V over $\mathbb{R}$, thus $\langle u + v, u - v \rangle = \langle u, u \rangle - \langle v, v \rangle = \|u\|^2 - \|v\|^2$.
- By $\uparrow$.
- We know 菱形 is a parallelogram that four sides have same length, thus $\|u\| = \|v\|$ and $u + v \perp u - v$, where $u + v$ and $u - v$ are two diagonal of 菱形.

□

Exercise 6.6. Let $u, v \in V$. Show that $\langle u, v \rangle = 0 \iff \|u\| \leq \|u + av\|$ for any $a \in F$.

Proof.

- ($\Rightarrow$) We will show that $\|u\| \leq \|u + av\|$ by $\|u\|^2 \leq \|u + av\|^2$ (recall that norm is always non-negative). $\|u + av\|^2 = \|u\|^2 + (a\|v\|)^2$

- ($\Leftarrow$) If $v = 0$, then $0 \perp u$ and the proof is complete, we assume $v \neq 0$. Let $w \in V$ such that $cv + w = u$ and $v \perp w$, then $\|u\|^2 = \|cv + w\|^2 = \|cv\|^2 + \|w\|^2$ (see ??), thus $\|u\|^2 \geq \|w\|^2$, therefore $\|u\| \geq \|w\|$ where $w = u - cv$, therefore $\|w\| \leq \|w\|$, hence $\|u\| = \|w\|$. Then $\|u\|^2 = \|cv\|^2 = \|w\|^2$ is now $\|cv\|^2 = 0$, therefore $c = 0$ or $v = 0$, but $v \neq 0$, thus $c = \frac{\langle u, v \rangle}{\|v\|^2} = 0$ then $\langle u, v \rangle = 0$ and $u \perp v$.

□

Exercise 6.7. Let $u, v \in V$. Show that $\|au + bv\| = \|bu + av\|$ for any $a, b \in \mathbb{R} \iff \|u\| = \|v\|$.

Proof.

- We have $\|au + bv\|^2 = \langle au, au \rangle + \langle bv, bv \rangle + \langle au, bv \rangle + \langle bv, au \rangle$ and $\|bu + av\|^2 = \langle bu, bu \rangle + \langle av, av \rangle + \langle bu, av \rangle + \langle av, bu \rangle$. Note that $\langle au, bv \rangle = a\bar{b}\langle u, v \rangle = \bar{a}b\langle u, v \rangle = \langle bu, av \rangle$, similarly, $\langle bv, au \rangle = \langle av, bu \rangle$. Thus $\langle au, au \rangle + \langle bv, bv \rangle = \langle bu, bu \rangle + \langle av, av \rangle$, let $a = \sqrt{2}$ and $b = 1$, we have $\sqrt{2}^2\langle u, u \rangle + \langle v, v \rangle = \langle u, u \rangle + \sqrt{2}^2\langle v, v \rangle$ therefore $\langle u, u \rangle = \langle v, v \rangle$, thus $\|u\| = \|v\|$.
- $\|au + bv\|^2 - \|bu + av\|^2 = a^2\langle u, u \rangle + b^2\langle v, v \rangle + b^2\langle u, u \rangle + a^2\langle v, v \rangle = 0$.

□

Exercise 6.9. Let $\|u\| = \|v\| = 1$ and $\langle u, v \rangle = 1$. Show that $u = v$.

Proof. We know $\langle u, v \rangle = \langle v, u \rangle = 1$ and then $\langle u, v \rangle - \langle v, v \rangle = 1 - 1 = 0 = \langle u - v, v \rangle$ and $-\langle v, u \rangle + \langle u, u \rangle = -1 + 1 = 0 = \langle u - v, u \rangle$ therefore $\langle u - v, v \rangle - \langle u - v, u \rangle = \langle u - v, u - v \rangle = 0$, thus $u - v = 0$. □

Exercise 6.10. Let $u, v \in V$ and $\|u\|, \|v\| \leq 1$. Show that:

$$\sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} \leq 1 - |\langle u, v \rangle|$$

Proof. By Cauchy-Schwarz inequality, we have $1 - |\langle u, v \rangle| \geq 1 - \|u\| \|v\|$. We

will show that $\sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} \leq 1 - \|u\| \|v\|$.

$$\begin{aligned}
& \sqrt{1 - \|u\|^2} \sqrt{1 - \|v\|^2} \leq 1 - \|u\| \|v\| \\
& (1 - \|u\|^2)(1 - \|v\|^2) \leq (1 - \|u\| \|v\|)^2 \\
& 1 - \|u\|^2 - \|v\|^2 + \|u\|^2 \|v\|^2 \leq 1 - 2 \|u\| \|v\| + \|u\|^2 \|v\|^2 \\
& -\|u\|^2 - \|v\|^2 \leq -2 \|u\| \|v\| \\
& \|u\|^2 + \|v\|^2 \geq 2 \|u\| \|v\| \\
& \|u\|^2 - 2 \|u\| \|v\| + \|v\|^2 \geq 0 \\
& (\|u\| - \|v\|)^2 \geq 0
\end{aligned}$$

The second step is well since both sides are non-negative. □

Exercise 6.11. Find $u, v \in \mathbb{R}^2$, such that u a scalar multiple of $(1, 3)$ and v is perpendicular to $(1, 3)$ and $(1, 2) = u + v$.

Proof. 人话：找到一个关于 $(1, 3)$ 和垂直于 $(1, 3)$ 的向量的线性组合，使得它等于 $(1, 2)$ 。

We have $(3, -1) \perp (1, 3)$, then $x = \frac{7}{10}$ and $y = \frac{1}{10}$ is a solution of:

$$\begin{aligned}
x + 3y &= 1 \\
3x - 1y &= 2
\end{aligned}$$

Thus $u = \frac{7}{10}(1, 3)$ and $v = \frac{1}{10}(3, -1)$. □

Exercise 6.14. Let $v \in V$ and $v \neq 0$. Show that for any $u \in V$ and $\|u\| = 1$, then

$$\left\| v - \frac{v}{\|v\|} \right\| \leq \|v - u\|$$

equal only if $u = \frac{v}{\|v\|}$.

这段话说的是： v 穿过单位圆的点的向量是距离 v 最近的元素。

Proof. Let $v' = \frac{v}{\|v\|}$ and $w \in V$ where $w \perp v'$ such that $u = cv' + w$, we can

see $|c| \leq 1$ by 勾股定理. Then

$$\begin{aligned}
\|v - v'\| &\leq \|v - u\| \\
\|v - v'\|^2 &\leq \|v - u\|^2 \\
\langle v, v \rangle + \langle -v', -v' \rangle + \langle v, -v' \rangle + \langle -v', v \rangle &\leq \langle v, v \rangle + \langle -u, -u \rangle + \langle v, -u \rangle + \langle -u, v \rangle \\
\langle v, -v' \rangle + \langle -v', v \rangle &\leq \langle v, -u \rangle + \langle -u, v \rangle \\
-\langle v, v' \rangle - \langle v', v \rangle &\leq -\langle v, u \rangle - \langle u, v \rangle \\
(-1)(\langle v, v' \rangle + \overline{\langle v, v' \rangle}) &\leq (-1)(\langle v, cv' + w \rangle + \langle cv' + w, v \rangle) \\
\langle v, v' \rangle + \overline{\langle v, v' \rangle} &\geq \langle v, cv' + w \rangle + \langle cv' + w, v \rangle \\
\langle v, v' \rangle + \langle v, v' \rangle &\geq \langle v, cv' \rangle + \langle v, w \rangle + \langle cv', v \rangle + \langle w, v \rangle \\
2\langle v, v' \rangle &\geq \langle v, cv' \rangle + \langle cv', v \rangle \\
2\langle v, v' \rangle &\geq \bar{c}\langle v, v' \rangle + c\langle v', v \rangle \\
2\langle v, v' \rangle &\geq \bar{c}\langle v, v' \rangle + c\overline{\langle v, v' \rangle} \\
2\langle v, v' \rangle &\geq \bar{c}\langle v, v' \rangle + c\langle v, v' \rangle \\
2 &\geq \bar{c} + c \\
2 &\geq 2 \operatorname{Re} c \\
1 &\geq \operatorname{Re} c
\end{aligned}$$

We have $\operatorname{Re} c \leq |c|$ and $|c| \leq 1$, thus $\operatorname{Re} c \leq 1$.

The inequality equals if $\operatorname{Re} c = 1$, which means $c = 1$ as $|c| = 1$ so the imagine part is 0, thus $u = cv' + w = v' + w$ while $\|u\| = \|v'\| = 1$, which means $w = 0$, therefore $u = v'$. $\square$

Exercise 6.15. Let non-zero $u, v \in \mathbb{R}^2$. Show that:

$$\langle u, v \rangle = \|u\| \|v\| \cos \theta$$

where θ is the angle between u and v .

Proof. Note that the inner product of $\mathbb{R}^2$ is euclidian inner product, then let

$u = a(\cos \alpha, \sin \alpha)$ and $v = b(\cos \beta, \sin \beta)$, thus

$$\begin{aligned}
\langle u, v \rangle &= ab(\cos \alpha \cos \beta) + ab(\sin \alpha \sin \beta) \\
&= ab(\cos \alpha \cos \beta + \sin \alpha \sin \beta) \\
&= ab(\cos \alpha \cos(-\beta) - \sin \alpha \sin(-\beta)) \\
&= ab(\cos(\alpha - \beta)) \\
&= \|u\| \|v\| (\cos(\alpha - \beta))
\end{aligned}$$

Recall that $\|u\| = a$ and $\|v\| = b$, since $\|(\cos \alpha, \sin \alpha)\| = 1$, similar to b . Also $\cos \theta = \cos -\theta$ and $\sin -\theta = -\sin \theta$. $\square$

Exercise 6.16. *The angle between two vectors in R^n are hard to see when $n > 3$, thus define the angle between $u, v \in R^n$ by:*

$$\arccos \frac{\langle x, y \rangle}{\|x\| \|y\|}$$

, try explain why this requires Cauchy-Schwarz inequality.

Proof. Note that $\arccos$ is the inverse of $\cos$, thus $\arccos$ only defines at $[-1, 1]$, which means $|\frac{\langle x, y \rangle}{\|x\| \|y\|}| \leq 1$, Cauchy-Schwarz inequality says $|\langle x, y \rangle| \leq \|x\| \|y\|$, thus $|\frac{\langle x, y \rangle}{\|x\| \|y\|}| \leq 1$. $\square$

Exercise 6.20. *Show that $u, v \in V$, then $|\|u\| - \|v\|| \leq \|u - v\|$.*

Proof. We will show $|\|u\| - \|v\||^2 \leq \|u - v\|^2$, we can see that no matter $\|u\| - \|v\|$ greater or less than 0, $|\|u\| - \|v\||^2$ is always $\|u\|^2 + \|v\|^2 - 2\|u\| \|v\|$, thus

$$\begin{aligned}
\|u\|^2 + \|v\|^2 - 2\|u\| \|v\| &\leq \langle u, u \rangle + \langle v, v \rangle + \langle u, -v \rangle + \langle -v, u \rangle \\
-2\|u\| \|v\| &\leq (-1)(\langle u, v \rangle + \langle v, u \rangle) \\
-2\|u\| \|v\| &\leq (-1)(\langle u, v \rangle + \overline{\langle u, v \rangle}) \\
-2\|u\| \|v\| &\leq (-2) \operatorname{Re} \langle u, v \rangle \\
\|u\| \|v\| &\geq \operatorname{Re} \langle u, v \rangle
\end{aligned}$$

Recall that $|\langle u, v \rangle| \leq \|u\| \|v\|$ and $\operatorname{Re} \langle u, v \rangle \leq |\operatorname{Re} \langle u, v \rangle| \leq |\langle u, v \rangle|$. $\square$

Exercise 6.22. Let $u, v \in V$, show that

$$\|u + v\| \|u - v\| \leq \|u\|^2 + \|v\|^2$$

Proof. The the left hand side is:

$$\begin{aligned} & \|u + v\| \|u - v\| \\ &= \sqrt{\langle u + v, u + v \rangle \langle u - v, u - v \rangle} \\ &= \sqrt{(\|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle)(\|u\|^2 + \|v\|^2 + \langle u, -v \rangle + \langle -v, u \rangle)} \\ &= \sqrt{(\|u\|^2 + \|v\|^2 + (\langle u, v \rangle + \langle v, u \rangle))(\|u\|^2 + \|v\|^2 - (\langle u, v \rangle + \langle v, u \rangle))} \end{aligned}$$

let $a = \|u\|^2 + \|v\|^2$ and $b = \langle u, v \rangle + \langle v, u \rangle$, then

$$\begin{aligned} & \|u + v\| \|u - v\| \\ &= \sqrt{(a + b)(a - b)} \\ &= \sqrt{a^2 - b^2} \end{aligned}$$

while the right hand side is: $\|u\|^2 + \|v\|^2 = a$, then square both sides we have:

$$a^2 - b^2 \leq a^2$$

which is obviously true, cause $b^2 \geq 0$ (it is not the case that b is a complex number, as $b = \|u + v\|^2 - \|u\|^2 - \|v\|^2$). $\square$